# 2D Crystallographic Point Groups using Symmetry-Breaking

Assignment 6: Bravais Lattice Tile Shading

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## 1 Introduction

The crystallographic restriction theorem tells us that a 2D periodic lattice can only be compatible with rotational symmetries of order $n \in \{1, 2, 3, 4, 6\}$. Combining these rotations with possible mirror reflections yields exactly **10 distinct 2D crystallographic point groups**.

The strategy used here mirrors the "dice analogy" from lectures: a fair die has the full octahedral symmetry, but adding ink dots breaks it down to lower point groups. We do exactly the same thing with geometric tiles placed at every Bravais Lattice (BL) point.

**Two base tiles are used:**

- **Square tile** – subdivided into 8 equal right-triangular sectors by cutting along both diagonals and both midlines. Sectors are labelled $s_1, s_2, \ldots, s_8$ clockwise starting from the upper-right.
- **Regular hexagonal tile** – subdivided into 12 equal triangular sectors by drawing lines from the centre to each of the 6 vertices *and* to each of the 6 edge midpoints. Sectors are labelled $h_1, h_2, \ldots, h_{12}$ clockwise from the top.

**Shading rule:** An unshaded tile represents the *highest* symmetry available to that family. Filling (darkening) one or more sectors lowers the symmetry, exactly as dots on a die break the cube's symmetry. The surviving symmetries are those rotations and reflections that map every shaded sector onto another shaded sector.

**Diagram legend:**

- Dark-filled triangle = shaded sector (marks asymmetry)
- Blue dashed line = mirror plane
- Small white shape at centre = rotation-axis marker (◯: 2-fold, □: 4-fold, △: 3-fold, : 6-fold)

*Note on Schoenflies vs. Hermann–Mauguin:* The Schoenflies symbol $C_n$ denotes a cyclic group of order $n$ (pure rotation), while $C_{nv}$ adds $n$ vertical mirror planes. In 2D the "vertical" planes are simply in-plane mirror lines.

# 2 Groups from the Square Tile (Oblique & Rectangular & Square Lattices)

The square tile naturally carries 4-fold rotational symmetry and 4 mirror lines (2 axial + 2 diagonal), giving a maximum point-group symmetry of $C_{4v}$ (HM: $4mm$). By progressively shading sectors we access all groups down to $C_1$.

## 2.1 Group $C_1$ (HM: 1) – No symmetry

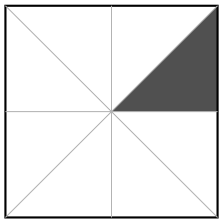

Figure 1: $C_1$: Only one sector ($s_2$) shaded. The tile has no rotational or mirror symmetry – only the trivial identity.

**Symmetry elements:** Identity $E$ only.
**Group order:** 1.
**Reasoning:** Sector $s_2$ lies in one quadrant of the square. No rotation by 90°, 180°, or 270° sends $s_2$ back to itself, and no mirror line maps the tile onto an identical-looking tile. This is the completely asymmetric case.

## 2.2 Group $C_s$ (HM: $m$) – One mirror, no rotation

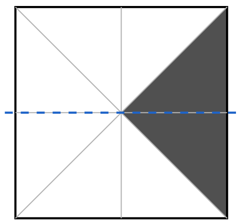

Figure 2: $C_s$: Sectors $s_2$ and $s_3$ shaded (right half, above & below midline). One horizontal mirror line preserved.

**Symmetry elements:** $E$, one mirror $\sigma$.
**Group order:** 2.
**Reasoning:** Together $s_2$ and $s_3$ fill the entire right side of the square between the two diagonals on the right. Reflecting across the vertical axis swaps $s_2 \leftrightarrow s_2$ and $s_3 \leftrightarrow s_3$ (each is its own mirror image across the horizontal midline), so the pattern is invariant. No 180° rotation works because the shading is asymmetric top-to-bottom overall (the left half is completely empty).

### 2.3 Group $C_2$ (HM: 2) – 2-fold rotation, no mirror

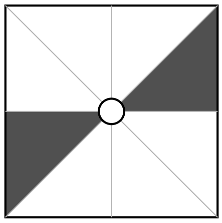

Figure 3: $C_2$: Sectors $s_2$ and $s_6$ shaded – diagonally opposite, producing a 2-fold (pinwheel) pattern.

**Symmetry elements:** $E$, $C_2$ (rotation by 180°).
**Group order:** 2.
**Reasoning:** Rotating 180° sends $s_2 \mapsto s_6$ and $s_6 \mapsto s_2$, so the pattern is invariant. No mirror line works: any candidate mirror would have to map $s_2$ to another shaded sector, but the only other shaded sector is $s_6$ and no single straight line through the centre relates these two triangles by reflection without also mapping unshaded sectors onto each other inconsistently.

### 2.4 Group $C_{2v}$ (HM: $2mm$) – 2-fold rotation + 2 mirrors

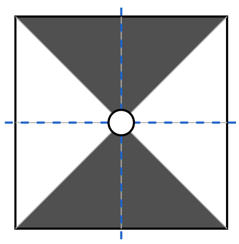

Figure 4: $C_{2v}$: Sectors $s_1, s_4, s_5, s_8$ shaded – the four "axial" triangles adjacent to the mid-lines. Two perpendicular mirrors and a 2-fold axis are present.

**Symmetry elements:** $E$, $C_2$, $\sigma_h$, $\sigma_v$.
**Group order:** 4.
**Reasoning:** The shaded sectors sit symmetrically about *both* the horizontal and vertical midlines. Reflecting across either midline maps each shaded sector to another shaded sector. The 180° rotation likewise permutes $s_1 \leftrightarrow s_5$ and $s_4 \leftrightarrow s_8$. This mimics the symmetry of a (non-square) rectangle.

### 2.5 Group $C_4$ (HM: 4) – 4-fold rotation, no mirror

**Symmetry elements:** $E$, $C_4$, $C_4^2 = C_2$, $C_4^3$.
**Group order:** 4.
**Reasoning:** Rotating by 90° sends $s_1 \to s_3 \to s_5 \to s_7 \to s_1$ (all shaded), so all four rotational powers of $C_4$ are symmetries. Any mirror line (horizontal, vertical, or diagonal) maps some shaded sector to an *unshaded* neighbour, breaking invariance. This is the "chiral square" – it has a handedness.

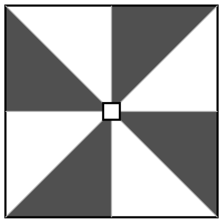

Figure 5: $C_4$: Every *other* sector shaded – a 4-bladed pinwheel. Rotation by 90° maps shaded→shaded, but any mirror line maps shaded→unshaded.

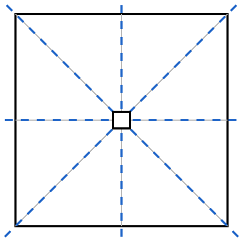

Figure 6: $C_{4v}$: Unshaded square – maximal symmetry. Four mirror lines (2 axial + 2 diagonal) and 4-fold rotation.

### 2.6 Group $C_{4v}$ (HM: $4mm$) – Full square symmetry

**Symmetry elements:** $E$, $2C_4$, $C_2$, $2\sigma_v$, $2\sigma_d$.
**Group order:** 8.
**Reasoning:** No sector is distinguished from any other; any rotation by a multiple of 90° and any of the four mirror lines maps the tile onto itself identically. This is the dihedral group $D_4$ (or $C_{4v}$ in Schoenflies), the full symmetry of a square.

## 3 Groups from the Hexagonal Tile (Hexagonal Lattice)

A regular hexagon has 6-fold rotational symmetry and 6 mirror lines, giving maximum $C_{6v}$ (HM: $6mm$). Shading selects sub-groups.

### 3.1 Group $C_3$ (HM: 3) – 3-fold rotation only

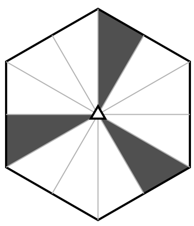

Figure 7: $C_3$: Three sectors spaced 120° apart (a 3-bladed pinwheel). Rotation by 120° maps blade to blade; mirrors break the pattern.

**Symmetry elements:** $E$, $C_3$, $C_3^2$.
**Group order:** 3.

**Reasoning:** Rotating by 120° cycles $h_1 \to h_5 \to h_9 \to h_1$ (all shaded). Reflecting across any line through the centre maps a shaded sector to an *adjacent* unshaded sector, destroying invariance. The pattern is a chiral 3-armed figure.

## 3.2 Group $C_{3v}$ (HM: $3m$) – 3-fold rotation + 3 mirrors

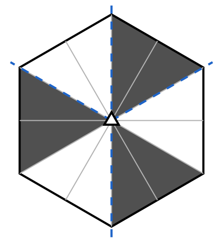

Figure 8: $C_{3v}$: Three adjacent-sector pairs shaded (arrowhead shapes), each pair bridging a vertex of the hexagon. Three mirror lines through each arrowhead axis.

**Symmetry elements:** $E$, $2C_3$, $3\sigma_v$.
**Group order:** 6.
**Reasoning:** Each shaded pair (e.g. $h_1+h_2$) forms a larger triangle pointing outward from a hexagon vertex. The mirror line through that vertex maps $h_1 \leftrightarrow h_2$ within the pair. Rotating 120° cycles each pair to the next. This is the symmetry of an equilateral triangle.

## 3.3 Group $C_6$ (HM: 6) – 6-fold rotation, no mirror

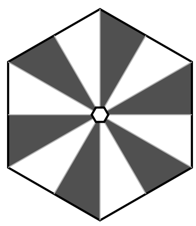

Figure 9: $C_6$: Every other sector shaded – a 6-bladed pinwheel. Each 60° rotation maps blade to blade; mirrors would swap blades with gaps.

**Symmetry elements:** $E$, $C_6$, $C_3$, $C_2$, $C_3^2$, $C_6^5$.
**Group order:** 6.
**Reasoning:** The alternating shading pattern (shaded / unshaded / shaded / ...) is preserved under any rotation that is a multiple of 60°, since each shaded sector steps to the next shaded sector. A reflection would map a shaded sector to an adjacent *unshaded* sector, ruining invariance. This is a chiral hexagonal pattern.

## 3.4 Group $C_{6v}$ (HM: $6mm$) – Full hexagonal symmetry

**Symmetry elements:** $E$, $2C_6$, $2C_3$, $C_2$, $3\sigma_v$, $3\sigma_d$.
**Group order:** 12.
**Reasoning:** An unmarked regular hexagon is invariant under any rotation by a multiple of 60° and under reflection across any of its 6 lines of symmetry (3 through opposite vertices, 3 through opposite edge midpoints). No sector is distinguished, so the full dihedral group $D_6$ is realised.

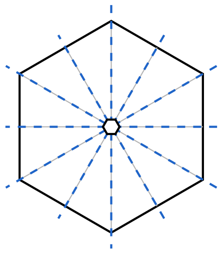

Figure 10: $C_{6v}$: Unshaded hexagon – maximal symmetry. Six mirror lines and 6-fold rotational axis.

# 4 Summary Table

| HM Symbol | Schoenflies | Tile | Shaded Sectors | Symmetry Retaine |
|---|---|---|---|---|
| 1 | $C_1$ | Square (8) | $s_2$ only | Identity only |
| $m$ | $C_s$ | Square (8) | $s_2, s_3$ | 1 mirror |
| 2 | $C_2$ | Square (8) | $s_2, s_6$ (opposite) | 180° rotation |
| $2mm$ | $C_{2v}$ | Square (8) | $s_1, s_4, s_5, s_8$ | 180° + 2 mirrors |
| 4 | $C_4$ | Square (8) | $s_1, s_3, s_5, s_7$ (alternate) | 90° rotation |
| $4mm$ | $C_{4v}$ | Square (8) | None (unshaded) | 90° + 4 mirrors |
| 3 | $C_3$ | Hexagon (12) | $h_1, h_5, h_9$ (120° apart) | 120° rotation |
| $3m$ | $C_{3v}$ | Hexagon (12) | $h_1h_2, h_5h_6, h_9h_{10}$ | 120° + 3 mirrors |
| 6 | $C_6$ | Hexagon (12) | $h_1, h_3, h_5, h_7, h_9, h_{11}$ (alternate) | 60° rotation |
| $6mm$ | $C_{6v}$ | Hexagon (12) | None (unshaded) | 60° + 6 mirrors |

Table 1: All ten 2D crystallographic point groups, their Schoenflies symbols, and the shading strategy used to realise each one from a maximally symmetric tile.